DGMS JSO GEOLOGY 2019 • GEOCHEMICAL EXPLORATION & ORE PROSPECTING

Question 100: Geochemical Anomalies in Mineral Exploration

Statistical Definitions of Background, Threshold, and Anomaly in Ore Discovery

Q.100: Which one of the following can be termed as significant geochemical anomaly in mineral exploration?

(a) Concentration of indicator elements lower than the local threshold

(b) Concentration of indicator elements falling within the regional threshold

(c) Concentration of indicator elements falling within the local threshold

✓ **(d) Concentration of indicator elements higher than the local background**

Correct Option: (d) Concentration of indicator elements higher than the local background

1. Fundamental Concepts in Exploration Geochemistry

In geochemical prospecting (Hawkes & Webb, 1962; Rose et al., 1979; Levinson, 1980), the primary objective is to differentiate between normal barren country rock and abnormal concentrations associated with hidden mineral deposits:

- **Geochemical Background:** The normal, baseline range of concentrations of an element in barren or unmineralized rocks, glacial tills, stream sediments, or

soils within a surveyed district. It represents natural statistical fluctuation around an average mean value (μ or $\bar{x}$).

- **Geochemical Threshold:** The upper limit of normal background variation. Conventionally defined as:

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Threshold = Mean Background (x̄) + 2 × Standard Deviation (s)
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Assuming a normal or log-normal distribution, only ~2.3% of unmineralized samples will naturally exceed this value.

- **Geochemical Anomaly:** Any statistically significant departure from the background. Specifically, an exploration anomaly is an abnormal concentration of indicator (pathfinder or target ore) elements that is **substantially higher than the local background and local threshold**, caused by primary or secondary dispersion halos surrounding an ore deposit.

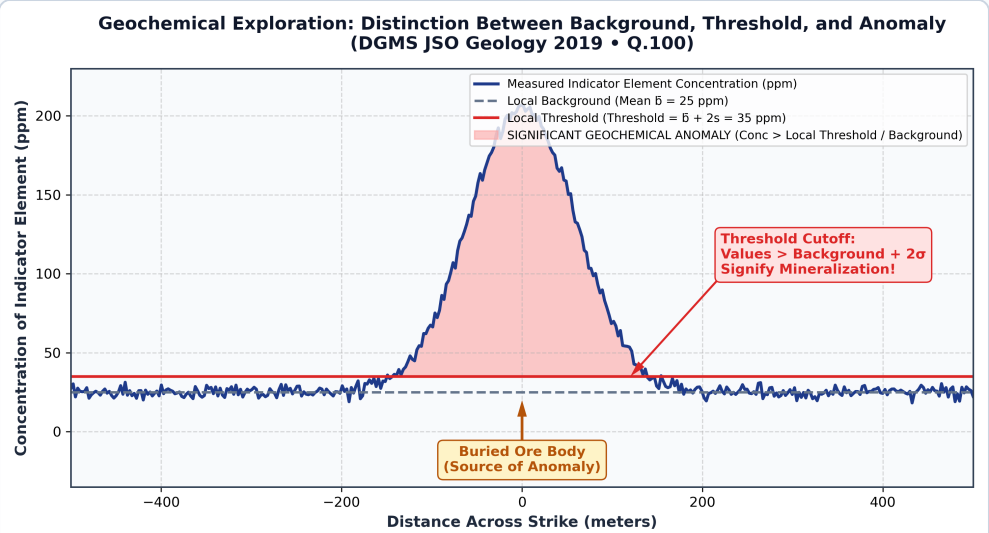


Figure 100.1: Typical geochemical dispersion traverse across a buried ore body, illustrating the relationship between baseline local background, statistical threshold cutoff (Mean + 2s), and a significant geochemical anomaly. Based on Hawkes & Webb, Geochemistry in Mineral Exploration.

2. Evaluating "Local Background" vs. "Local Threshold"

Because natural rocks vary greatly between lithologies (e.g., ultramafic vs. granitic terrains), background levels must be evaluated locally. A **significant geochemical**

anomaly is identified when element concentrations rise prominently above the local background baseline:

- Values that fall *within* or *lower than* the local threshold represent ordinary barren background variations.
- Values that rise **higher than the local background** and surpass the threshold define true anomalies that justify expensive follow-up stages such as trenching, geophysical sounding, and diamond-core drilling.

3. Systematic Option Evaluation

Option	Geochemical Concentration Condition	Exploration Significance	Anomaly Status
(a)	Concentration of indicator elements lower than the local threshold	Represents ordinary unmineralized background rock or geochemical depletion zones.	Not an anomaly
(b)	Concentration of indicator elements falling within the regional threshold	Represents normal broad regional background fluctuation across unmineralized terranes.	Not an anomaly
(c)	Concentration of indicator elements falling within the local threshold	Represents expected random local background noise in barren host formations.	Not an anomaly
(d)	**Concentration of indicator elements higher than the local background**	**Signifies an abnormal accumulation of indicator elements (e.g., Cu, Pb, Zn, Au, As) well above the local barren baseline, indicating primary or secondary dispersion**	**Significant Geochemical Anomaly**

Option	Geochemical Concentration Condition	Exploration Significance	Anomaly Status
		halos of an ore deposit.	

> **Key Exploration Takeaway**
>
> In geochemical exploration, an **anomaly** is fundamentally defined as a concentration of target or pathfinder elements **higher than the local background**. When this concentration exceeds the threshold (typically $\bar{x} + 2s$), it is classified as a significant anomaly indicating proximity to hidden economic mineralization.

4. Standard Textbook References

- **H. E. Hawkes & J. S. Webb**, *Geochemistry in Mineral Exploration* (Harper & Row, 1962), Chapter 2: "Principles of Geochemical Dispersion", Section: Background, Threshold, and Anomalies, pp. 23–45.
- **A. W. Rose, H. E. Hawkes, & J. S. Webb**, *Geochemistry in Mineral Exploration* (2nd Edition, Academic Press, 1979), Chapter 3: "Statistical Interpretation of Geochemical Data", pp. 31–56.
- **A. A. Levinson**, *Introduction to Exploration Geochemistry* (2nd Edition, Applied Publishing Ltd., 1980), Chapter 6: "The Geochemical Anomaly", pp. 195–230.